

BABAPPA: a research-alpha, simulation-trained framework for guarded branch-site positive-selection support under alignment uncertainty

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Abstract

Branch-site positive-selection analysis is sensitive to multiple sequence alignment, taxon sampling, saturation, hidden paralogy, and model misspecification. BABAPPA (Branch-site Alignment-Bias-Aware Probabilistic Positive-selection Analyzer) is a research-alpha software and methods framework designed to provide guarded branch-site positive-selection support rather than unsupported empirical discovery calls. The framework combines simulation-supervised explicit branch-site truth, a production alignment ensemble, conservative branch-site features, tier-aware neural scoring, calibration and aggregation layers, deployable model packaging, empirical input quality control, out-of-distribution (OOD) applicability gates, and reference-tool comparison scaffolding.

The current implementation, 0.5.2-alpha, completed a conservative explicit branch-truth Apple Silicon/MPS validation at 100,000 simulated families. The validation used four saturation tiers, explicit simulator branch-site truth, the `conservative_branch_site` feature policy, and the production alignment ensemble `identity`, MAFFT, BABAPPAalign, and MUSCLE. PRANK and T-Coffee are retained only as diagnostic aligners. The 100K run is a conditional pass: raw and intermediate directories were intentionally pruned after completion, but retained summaries, truth audits, model artifacts, calibration outputs, aggregation outputs, controls, and stage markers support the completed validation state. Branch-site neural test AUROC was 0.998605, 0.996962, 0.994527, and 0.989955 for low, moderate, high, and extreme saturation tiers. Branch-level AUROC was 0.999797, 0.999421, 0.998777, and 0.997873. Branch-to-gene AUROC was 0.999697, 0.999616, 0.999508, and 0.999430. These are simulation-supervised validation results only.

The project also includes a validated deployable simulation-trained package for the conservative 100K MPS model family, plus an empirical bridge for input validation, alignment-ensemble construction, feature extraction, truth-column auditing, applicability/OOD checking, branch-site scoring, empirical reporting, codeml/HyPhy-style comparison, and simulation-matched calibration planning. A WRKY empirical pilot demonstrates both abstention and guarded scoring. `WRKY_candidate_01` was correctly treated as an OOD stress test. `WRKY_candidate_02_close` produced a BABAPPA diagnostic-positive signal on close-taxa WRKY33/AT2G38470 homologs, but codeml and HyPhy references were negative. A 100-replicate feature-level matched-null calibration was completed and showed unusual called-row burden but not unusual maximum gene support. Therefore the correct interpretation is BABAPPA-only, mixed feature-level null support, and inconclusive as an empirical discovery claim. BABAPPA is ready as a guarded research-alpha software/methods contribution and for calibrated empirical expansion, but not as a finalized empirical positive-selection discovery engine.

1 Introduction

Branch-site positive-selection inference is a central task in molecular evolution. Classical codon models estimate lineage- and site-specific shifts in the nonsynonymous-to-synonymous rate ratio, $\omega = d_N/d_S$, and tools such as PAML/codeml and HyPhy have become standard references for testing episodic diversifying selection [4, 6, 7, 9–11, 14, 17, 18]. These methods remain essential. However, they usually condition on a single multiple sequence alignment and a fixed phylogenetic representation. Alignment errors, saturation, recombination, synonymous-rate variation, multinucleotide substitutions, hidden paralogy, and taxon-sampling problems can all distort downstream selection analyses [1–3, 5, 8, 12, 13, 15, 16].

BABAPPA addresses this problem from a software/methods perspective. It does not replace codeml or HyPhy, and it does not claim that a simulation-trained classifier can by itself prove empirical positive selection. Instead, BABAPPA formalizes a guarded diagnostic workflow. It trains on explicit simulator branch-site truth, confronts multiple alignment hypotheses, uses conservative branch-site features, records leakage and context-shortcut risks, gates empirical inputs by applicability/OOD criteria, and reports diagnostic scores that must be interpreted with matched-null calibration and external reference tools.

This manuscript has three goals. First, it describes the scientific design of BABAPPA as a simulation-trained, alignment-ensemble framework for branch-site support. Second, it reports the completed 100K explicit branch-truth validation and the deployable model package produced from it. Third, it documents the present empirical bridge and WRKY pilot without overstating biological claims. The correct status is research-alpha: suitable for software/methods communication and reproducible evaluation, not for unsupported empirical discovery claims.

2 Design principles

2.1 Simulator truth and empirical inference are separate

BABAPPA distinguishes simulator truth from empirical data. During simulation validation, explicit branch-site labels are available from the generative process and are used as supervised targets. During empirical inference, no simulator truth exists and no truth-derived columns are allowed as inputs. The empirical bridge therefore includes a truth-column audit that rejects or flags columns such as `branch_site_truth`, `selected_sites`, `truth`, `branch_truth`, `oracle`, `y_branch_site`, `y_site`, and `gene_label`. This separation is central to the claim boundary of the project.

2.2 Alignment ensembles are a core design element

Alignment uncertainty is not treated as a cosmetic nuisance. BABAPPA uses multiple alignment workflows so that branch-site features can be evaluated against alternative homology hypotheses. The current production/default aligner set is `identity`, `MAFFT`, `BABAPPAalign`, and `MUSCLE`. `PRANK` and `T-Coffee` remain diagnostic-only. Method-policy filtering and quarantine logic are used to record aligner failures or suspicious outputs rather than silently dropping them.

2.3 Conservative branch-site features are the deployable policy

The deployable branch-site workflow uses `conservative_branch_site`. This feature policy was chosen after foreground-context ablation showed shortcut risk from context-only information. The policy prioritizes features that support branch-site scoring while reducing direct leakage from foreground-context identifiers and simulator truth. This does not remove all empirical uncertainty; it defines the safest current research-alpha feature surface.

2.4 Applicability and abstention are first-class outputs

For empirical data, BABAPPA estimates whether an input family is in-domain, borderline, or out-of-domain relative to the training envelope. It considers quantities such as taxon count, codon count, pairwise divergence, saturation proxy, gap and ambiguity burden, site-map mapability, feature missingness, and alignment disagreement. OOD cases are diagnostic-only and must not be interpreted as positive-selection calls.

3 Software components

Table 1 summarizes the current executable components. The project is implemented as a command-line Python package with explicit validators, machine-readable JSON/TSV outputs, and Markdown reports.

Table 1: BABAPPA components and workflow roles.

Component	Role
Simulation and validation	Generate simulated coding families, record explicit branch-site truth, and validate simulation completeness.
Alignment ensemble	Run <code>identity</code> , <code>MAFFT</code> , <code>BABAPPAalign</code> , and <code>MUSCLE</code> as the production ensemble; retain <code>PRANK</code> and <code>T-Coffee</code> only for diagnostic stress tests.
Site maps and method policy	Map aligned coordinates to original coordinates, validate equal-length/codon-aware outputs, and quarantine problematic methods.
Tensor/dataset builders	Construct feature tables and family-disjoint splits with streaming and capping for large branch-site datasets.
Branch-site neural scoring	Train and validate conservative branch-site neural models on explicit simulator truth.
Calibration and aggregation	Calibrate branch-site scores, profile thresholds, aggregate to branch and gene support, and run aggregation controls.
Deployable package	Package tier models, calibrations, feature schemas, training envelope, checksums, validation summaries, and model-card limitations.

Empirical bridge	Validate empirical inputs, run empirical alignments, extract features, audit forbidden columns, score, and report with applicability/OOD status.
Reference comparison	Prepare and parse codeml/HyPhy-style reference workflows and compare BABAPPA diagnostic outputs to external results.
Maintenance	Audit storage with <code>babappa audit-storage</code> , classify reproducible heavy outputs, and generate safe quarantine scripts.

4 Validation progression

BABAPPA progressed through increasingly explicit and conservative validation stages. Early gene-level experiments showed that direct gene-level classification was unstable under saturation and could compress probabilities. Site-level and branch-site formulations were then developed to localize sparse simulator truth before aggregating to higher levels. The explicit branch-truth pathway progressed from 1K to 10K and then to 100K, with Apple Silicon/MPS support added for the user’s MacBook Pro M5 Max environment.

The completed 100K MPS validation is the current primary validation result. It used:

- truth mode: explicit simulator branch-site truth;
- feature policy: `conservative_branch_site`;
- methods: `identity`, `MAFFT`, `BABAPPAlign`, and `MUSCLE`;
- saturation tiers: low, moderate, high, and extreme;
- 25,000 simulated families per tier;
- streamed/capped branch-site datasets;
- Apple Silicon/MPS neural execution with conservative memory safeguards.

The final 100K decision is a conditional pass. The qualification is practical rather than a failure of retained validation: raw and intermediate directories were pruned after completion to recover storage, so they cannot be directly revalidated in place. The retained summaries, truth audits, model artifacts, calibration outputs, aggregation outputs, controls, and stage markers support completion. This condition is recorded in the final validation report and should remain visible whenever the 100K result is cited.

5 100K simulation-supervised validation results

Table 2 reports the retained 100K cross-tier performance values. These metrics evaluate supervised recovery of explicit simulator branch-site truth and derived aggregation targets. They are not empirical positive-selection performance estimates.

Table 2: Conservative explicit branch-truth 100K Apple Silicon/MPS validation metrics.

Saturation tier	Branch-site neural AUROC	Branch AUROC	Gene AUROC
Low	0.998605	0.999797	0.999697
Moderate	0.996962	0.999421	0.999616
High	0.994527	0.998777	0.999508
Extreme	0.989955	0.997873	0.999430

Performance degraded smoothly from low to extreme saturation rather than collapsing, which supports the feasibility of the conservative branch-site information pathway under the current simulator. At the same time, these values are high because the task is simulation-supervised and explicit truth is known. They should not be translated into empirical false-discovery rates without matched-null calibration and reference comparisons.

6 Deployable simulation-trained package

The validated deployable package is:

`deployable_model_conservative_branch_site_100k_mps`

Package validation status is `ok`, with zero failures and zero warnings. The package contains model manifests, model card, feature and calibration schemas, training envelope, tier models, tier calibrations, checksums, validation summary, limitations, and README files. The package excludes raw simulator truth, raw oracle labels, raw branch-site datasets, simulations, and alignments.

This package is suitable for guarded diagnostic empirical scoring only. It is simulation-trained, explicitly conservative, and should be used with empirical input validation, applicability/OOD checks, reference-tool comparison, biological controls, and simulation-matched calibration. It is not a final empirical inference product.

7 Empirical bridge

The empirical bridge implements the following guarded workflow:

1. empirical CDS FASTA and tree validation;
2. alignment ensemble construction using the production/default method set;
3. empirical site-map and method-policy generation;
4. branch-site feature extraction against the deployable feature schema;
5. forbidden truth-column audit;

6. applicability/OOD classification;
7. deployable-model branch-site scoring;
8. empirical report generation;
9. codeml/HyPhy reference workflow preparation, parsing, and comparison;
10. simulation-matched calibration planning and a conservative `run-simulation-matched-null-calibration` evidence-pack interface for dry-run validation and feature-level matched-null scoring.

This bridge is a software success, but it is not equivalent to biological discovery. It creates the infrastructure for responsible empirical analysis and makes the claim boundary explicit in every report. At present, the evidence-pack matched-null runner supports safe dry-run validation and conservative feature-level null scoring through the deployable model package. This is useful diagnostic calibration support, but it is not the same as a full raw sequence simulation plus alignment replay and should not be treated as a standalone empirical discovery criterion.

8 WRKY empirical pilot

Two WRKY-family empirical pilots demonstrate the current behavior of the empirical bridge (Table 3). They should be read as diagnostic software evaluations.

Table 3: WRKY empirical pilot outcomes and reference-tool comparison.

Family	BABAPPA status	Interpretation
WRKY_candidate_01	OOD stress test; mean p-distance 0.725799; applicability <code>out_of_domain</code> ; <code>diagnostic_only=True</code> ; no positive call.	Useful failure-mode probe showing that highly divergent WRKY panels can be rejected or treated as diagnostic-only. Not a biological result.

WRKY_candidate_02 _close	Close-taxa WRKY33/AT2G38470 ho- mologs; 7 Brassicaceae/close plant taxa; aligned mean p-distance approximately 0.101293; prefilter <code>accept</code> ; ap- plicability <code>in_domain</code> ; moderate tier; MPS scoring; diagnostic- positive; max gene support 0.177189; 6,954 called branch- site rows; 10,451 score rows; <code>diagnostic_only=False</code> .	Demonstrates that the em- pirical bridge can produce gated diagnostic scores on plausible biological inputs. It is not a positive-selection discovery claim.
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Reference workflows have also been run or parsed for `WRKY_candidate_02_close`. `codeml` branch-site Model A versus null gave LRT 0.0 and $p = 1.0$, negative. `HyPhy aBSREL` gave foreground $p = 1.0$, negative. `HyPhy MEME` gave a minimum $p = 0.0641705$, negative at 0.05. The concordance class is therefore `BABAPPA_only`. The completed 100-replicate feature-level matched-null calibration gave `p_empirical_called_rows=0.009900990099009901`, but `p_empirical_support=1.0` and `p_empirical_branch_support=1.0`.

The correct conclusion is: the WRKY close-taxa pilot demonstrates that the empirical bridge can produce gated diagnostic scores on plausible biological inputs; however, the `BABAPPA_only` positive signal is inconclusive as a discovery claim because `codeml` and `HyPhy` did not support positive selection and the null result is feature-level and mixed rather than a full raw sequence simulation/alignment replay.

9 Software maintenance and reproducibility

Version 0.5.2-alpha also includes `babappa audit-storage`, a storage-audit command that inventories a `BABAPPA` workspace, classifies protected artifacts, identifies reproducible generated outputs, and writes user-run quarantine and validation scripts. This is a reproducibility feature rather than a scientific result: heavy simulation and alignment intermediates can be pruned after final summaries, audits, checksums, model artifacts, and manifests are preserved.

The current release archive label is 0.5.0-alpha. The Zenodo-ready archive is recorded as:

```
BABAPPA_0.5.0-alpha_release_zenodo_20260530.tar.xz
SHA256: cc259617f19d9634fd6e11906903910498ab78d3797a10df1bb24b7db014dc30
```

The source version and release archive label differ because cleanup and maintenance tooling advanced after the archive label was established. Future releases should refresh both together.

10 Limitations

`BABAPPA` has several explicit limitations.

- The deployable model is simulation-trained. A simulator-to-real gap is unavoidable and must be tested.
- Empirical calibration is pending. BABAPPA diagnostic scores do not yet provide calibrated empirical p -values or final false-discovery guarantees.
- Empirical discovery claims are not finalized. A diagnostic-positive result is not, by itself, a publishable positive-selection claim.
- Alignment and taxon sampling remain important. Very high divergence, poor mappability, hidden paralogy, and taxon-mismatched trees can invalidate interpretation.
- External reference disagreement must be treated conservatively. BABAPPA-only signals should trigger matched-null calibration and controls, not discovery language.
- The context-only shortcut warning remains scientifically relevant. The conservative feature policy reduces but does not abolish the need for ablation and leakage audits.
- Raw/intermediate pruning limits direct reruns of some validators. The 100K validation is supported by retained summaries and artifacts, but the pruned raw directories cannot be revalidated in place.
- BABAPPA should be used alongside codeml, HyPhy, biological controls, and dataset-specific justification. It is not a replacement for reference workflows at this stage.

11 Conclusion

BABAPPA provides a validated research-alpha framework for guarded, simulation-trained branch-site positive-selection support and empirical diagnostic scoring. Its completed 100K explicit branch-truth MPS validation, deployable conservative model package, alignment-ensemble workflow, empirical OOD gate, and WRKY diagnostic pilot make it ready for software/methods communication and reproducible evaluation. It is not ready for unsupported empirical positive-selection discovery claims. The next scientific step is to complete simulation-matched null calibration, add close-taxa negative controls, and interpret BABAPPA outputs jointly with codeml/HyPhy-style references before any empirical claim is made.

Author contributions

Krishnendu Sinha conceived the project, designed the framework, implemented and validated the research-alpha workflow with AI-assisted software-development support, curated the empirical pilot state, and wrote the manuscript.

Competing interests

The author declares no competing interests.

Data and code availability

The code repository will be deposited at [PLACEHOLDER_REPOSITORY_URL]. The release archive label is 0.5.0-alpha and the current source version described here is 0.5.2-alpha. The archive filename and SHA256 checksum are reported in the reproducibility section above and should be replaced by a DOI-resolved archive at [PLACEHOLDER_ZENODO_DOI] when the formal release is deposited. Large raw simulations, alignments, tensor shards, and streamed datasets are not intended to be version-controlled; final summaries, audits, model manifests, checksums, validation reports, and evidence packs are the retained reproducibility artifacts.

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